Tutorials for the course BIE-MA1

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February 5, 2025

Tutorial n. 0

Basic notions

Basic equations and inequalities, set theory, elementary functions, basic properties of functions.

Notation

$\mathbb{N}$ .... natural numbers
$\mathbb{N}_0$ .... natural numbers plus 0, i.e. $\mathbb{N}_0$=$\mathbb{N} \cup \{0\}$
$\mathbb{Z}$ .... integer numbers
$\mathbb{R}$ .... real numbers
$A \cup B$ .... union of sets A and B
$A \cap B$ .... intersection of sets A and B
$A \setminus B$ .... subtraction of sets A and B
$A \times B$ .... Cartesian product of sets A and B
$\lceil x \rceil$ .... upper integer part of the real number x, $\emptyset$ .... empty set

Exercise 0.1: Let

$$A = \{x \in \mathbb{R} \mid -x^2 + 5x - 5 \geq 1\}.$$

and

$$B = \{x \in \mathbb{R} \mid 2 < x \leq 3\}.$$

Find $A \cup B$, $A \cap B$, $A \backslash B$, $A \backslash \mathbb{Z}$, $\mathbb{R} \backslash A$. Represent $A \times B$ and $B \times A$ in the Cartesian plane.

Solution: $A \cup B = A$, $A \cap B = B$, $A \backslash B = \{2\}$, $A \backslash \mathbb{Z} = (2,3)$, $\mathbb{R} \backslash A = (-\infty, 2) \cup (3, +\infty)$, ...

Exercise 0.2: Find the sets

$$A = \left\{ x \in \mathbb{R} \mid \left(\frac{1}{3}\right)^{(1-12x)x} < 3 \right\},$$

$\mathbb{R} \backslash A$ and $\mathbb{N} \backslash A$.

Solution: $A = (-\frac{1}{4}, \frac{1}{3})$, $\mathbb{R} \backslash A = (-\infty, -\frac{1}{4}] \cup [\frac{1}{3}, +\infty)$, $\mathbb{N} \backslash A = \mathbb{N}$.

Solution. The set A is the set of solutions of the inequality

$$\left(\frac{1}{3}\right)^{(1-12x)x} < 3 \tag{1}$$

over the set of real numbers. This is an inequality with exponential of base $\frac{1}{3}$ that is smaller than 1. Its solution is therefore given by the solution of the inequality

$$\log_{\frac{1}{3}} \left(\frac{1}{3}\right)^{(1-12x)x} > \log_{\frac{1}{3}} 3,$$

that simplifies to

$$x - 12x^2 > -1,$$
$$12x^2 - x - 1 < 0 \tag{2}$$

The equation $12x^2 - x - 1 = 0$ is solved by

$$x_{1,2} = \frac{1 \pm \sqrt{1+48}}{24},$$

thus $x_1 = -\frac{1}{4}$ and $x_2 = \frac{1}{3}$. Equation (2) is therefore solved by

$$x \in \left(-\frac{1}{4}, \frac{1}{3}\right),$$

that gives

$$A = \left(-\frac{1}{4}, \frac{1}{3}\right).$$

It follows that

$$\mathbb{R} \backslash A = \{x \mid x \in \mathbb{R} \text{ and } x \notin A\} = (-\infty, -\frac{1}{4}] \cup [\frac{1}{3}, +\infty).$$

Similarly, we find $\mathbb{N} \backslash A = \mathbb{N}$.
Note that we invert the inequality in (1) because the base $\frac{1}{3}$ is smaller than 1, otherwise we would not.

Exercise 0.3: Find for which values of $x \in \mathbb{R}$ the following inequalities are true:

a) $\left(\frac{1}{4}\right)^{x^2} > 2$

b) $3^{x^2} > 3$

c) $2^{x^2-4x+3} \geq 1$

d) $\left(\frac{1}{2}\right)^{x^2-4x+3} > 1$

e) $\left(\frac{1}{3}\right)^{1-x^2} \geq 0$

Solution: a) no solution in $\mathbb{R}$, b) $x < -1$ or $x > 1$, c) $x \leq 1$ or $x \geq 3$, d) $1 < x < 3$, e) $\forall x \in \mathbb{R}$.

Exercise 0.4: Let

$$B = \{x \in \mathbb{R} \mid |x - 6| \leq 3\}$$

Find $\mathbb{Z} \backslash B$, $A \cup B$ and $A \cap B$, where A is the set defined in exercise 0.2.
Solution: $\mathbb{Z} \backslash B = \{X \in \mathbb{Z} \mid x \leq 2 \text{ or } x \geq 10\}$, $A \cup B = (-\frac{1}{4}, \frac{1}{3}) \cup [3, 9]$ and $A \cap B = \emptyset$.

Solution. The inequality $|x - 6| \leq 3$ is solved by

$$\begin{aligned} -3 \leq x - 6 \leq 3, \\ 3 \leq x \leq 9, \end{aligned} \tag{3}$$

thus $B = [3, 9]$, $\mathbb{Z} \backslash B = \{x \in \mathbb{Z} \mid x \leq 2 \text{ or } x \geq 10\}$, $A \cup B = (-\frac{1}{4}, \frac{1}{3}) \cup [3, 9]$ and $A \cap B = \emptyset$.

Exercise 0.5: Given that $y = \log_a x$, prove that

a) $\log_{\frac{1}{a}} x = -y$

b) $\log_{ay} x = \frac{y}{1+y}$

c) $\log_x (ax)^y = 1 + y$

Hint: use the properties of the logarithm and its definition as inverse of the exponentiation.

Exercise 0.6: Find for which values of $x \in \mathbb{R}$ the following inequalities are true:

a) $\log x \geq 0$

b) $\ln x \leq -1$

c) $\ln(x^2 - 5x + 7) > 0$

d) $\ln(4 - x^2) \leq 0$

Solution. a) $x \geq 1$, b) $0 < x \leq \frac{1}{e}$, c) $x \in (-\infty, 2) \cup (3, +\infty)$, d) $x \in (-2, -\sqrt{3}] \cup [\sqrt{3}, 2)$.

Exercise 0.7: Let

$$A = \{x \in (0, +\infty) \ : \ \log x \leq 2\}.$$

Decide whether the set A is bounded (below and/or above) and find infimum, supremum, minimum and maximum for A, if they exist. Solution. $A = (0, 100]$, thus A is bounded. 0 is infimum, 2 is maximum (and supremum), mimimum does not exists.

Exercise 0.8: Solve for $x \in \mathbb{R}$

a) $\cos x \leq \frac{1}{2}$

b) $\cos x + \sin x \tan x > 1$

c) $\tan x \geq 1$

Solution: a) $\frac{\pi}{3} + 2\kappa\pi \leq x \leq \frac{5}{3}\pi + 2\kappa\pi$, $\kappa \in \mathbb{Z}$, b) $-\frac{\pi}{2} + 2\kappa\pi < x < \frac{\pi}{2} + 2\kappa\pi$ and $x \neq \kappa\pi$, $\kappa \in \mathbb{Z}$, c) $\frac{\pi}{4} + \kappa\pi \leq x < \frac{\pi}{2} + \kappa\pi$, $\kappa \in \mathbb{Z}$

Exercise 0.9: Consider the set

$$A = \{x \in [0, +\infty) \ : \ \sqrt[4]{x} \geq 2\}.$$

Decide whether the set A is bounded. Find, if they exists, minimum, maximum, infimum, supremum. Solution: $A = [16, +\infty)$. Thus A is bounded below, but not above, so A is not bounded. 16 is mimimum (and infimum), but there exist no supremum nor maximum.

Exercise 0.10: Find the sets $A \cup B$, $A \cap B$, $B \setminus A$, where A is the subset of real numbers given by the solutions of the inequality

$$\sqrt{\frac{x^2 + x - 2}{x - 5}} \leq 1$$

and B is the subset of real numbers given by the solutions of the equation

$$\sqrt{x^4 + 2x^2 + 1} = -3.$$

$A \cap B = \emptyset$, $A \cup B = A = [-2, 1]$, $B \setminus A = \emptyset$.

Solution. First let us solve
$$\sqrt{\frac{x^2+x-2}{x-5}} \leq 1.$$

This is an irrational inequality. Its solution is given by the set of values of x such that the following conditions are both satisfied:

$$\frac{x^2+x-2}{x-5} \geq 0 \text{ (i.e. the square root is well defined),} \tag{4}$$
$$\frac{x^2+x-2}{x-5} \leq 1 \tag{5}$$

To solve (4), we study the sign of numerator and denominator separately, and then we use the result so obtained to determine the sign of the whole fraction. For the numerator we have

$$x^2+x-2 \geq 0 \Leftrightarrow x \in (-\infty,-2] \cup [1,+\infty)$$

and for the denominator
$$x-5>0 \Leftrightarrow x>5.$$

The condition (4) is satisfied when numerator and denominator have the same sign or when the numerator equals 0, thus for $x \in [-2,1]$. Let us now solve (5). We can have two cases:

$$x^2+x-2 \leq x-5 \text{ and } x-5>0$$

or
$$x^2+x-2 \geq x-5 \text{ and } x-5<0.$$

In the first case we reduce to
$$x^2+3 \leq 0 \text{ and } x>5$$

that has no solution as we can not have at the same time $x = \pm\sqrt{3}$ and $x > 5$. In the second case we have
$$x^2+3 \geq 0 \text{ and } x<5$$

that it satisfied $\forall\, x < 5$. Thus (5) is solved by $x < 5$. Taking the intersection between the solution set of (4) and the solution set of (5) we find $A = [-2,1]$. To find B we consider the equation
$$\sqrt{x^4+2x^2+1} = -3.$$

This equation has no solution, since the square root operation always returns a positive value. Therefore we conclude that $A = [-2,1]$, $B = \emptyset$ and $A \cap B = \emptyset$, $A \cup B = A = [-2,1]$, $B \setminus A = \emptyset$.

Exercise 0.11: Solve for $x \in \mathbb{R}$

a) $2x^4 - x^2 + 1 < 0$

b) $x^{50} + 1 > 0$

Solution: a) no solution, b) $\forall\, x \in \mathbb{R}$.

Exercise 0.12: Decide whether the following set is bounded. If they exists find infimum, supremum, minimum and maximum of the set.

$$A = \{x \in \mathbb{R} \ : \ x^4 - x^2 - 12 > 0\}.$$

Solution: $A = (-\infty, -2) \cup (2, +\infty)$, thus A is has no infimum nor supremum, and therefore no minimum nor maximum and it not bounded.

Exercise 0.13: Solve for $x \in \mathbb{R}$

$$\frac{x^4 + 1}{x^3 + 6x^2 + 12x + 1} \geq 0$$

Exercise 0.14: Solve for $x \in \mathbb{R}$

$$\frac{x^6 + 1}{4x^4 - 4x^2 + 1} \geq 0$$